How journals can improve code and data sharing in Neuroscience

Stephen J Eglen
https://sje30.github.io
sje30@cam.ac.uk

Cambridge Computational Biology Institute
University of Cambridge
@StephenEglen

Slides: http://bit.ly/eglen-neuromatch (CC-BY license)

Declarations

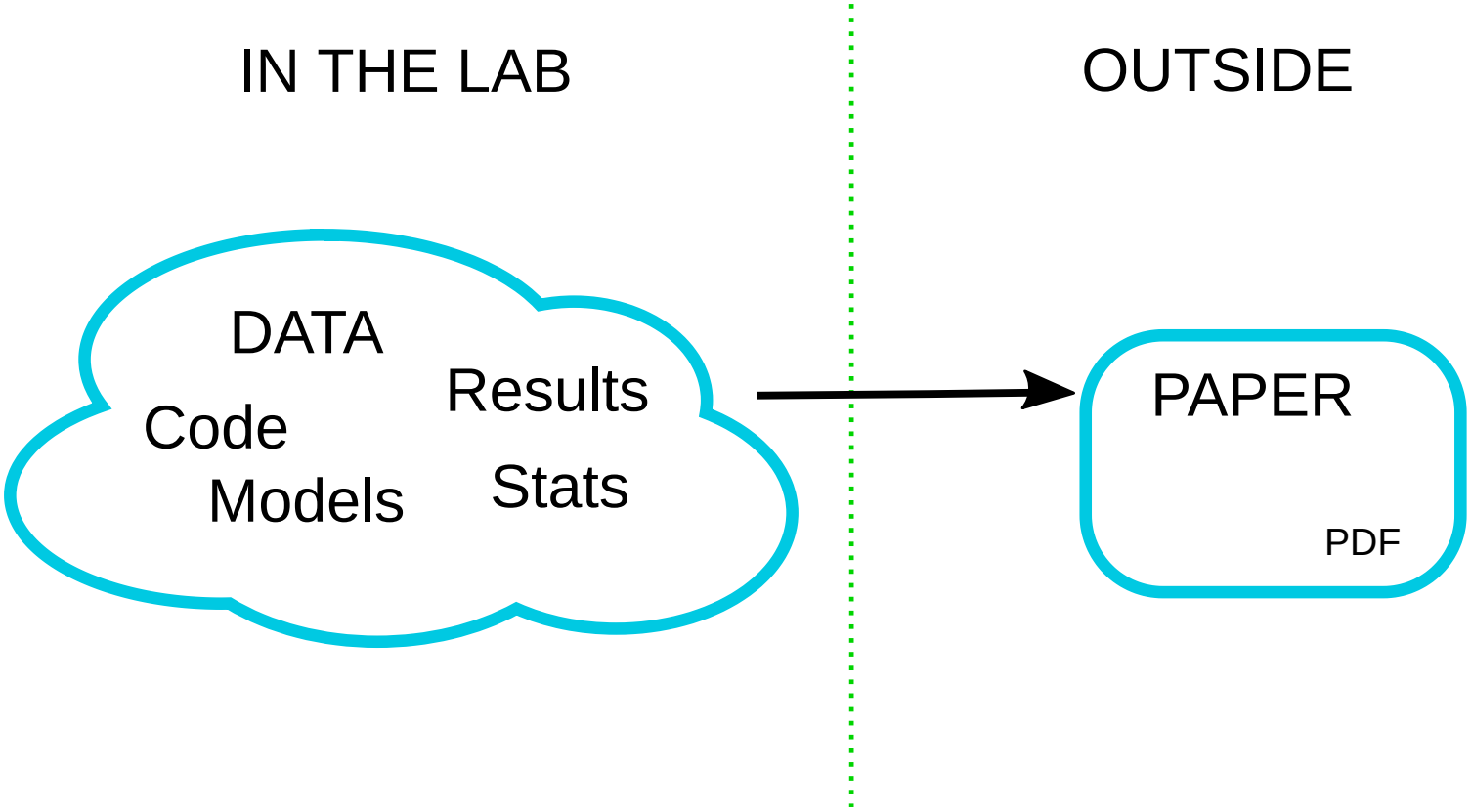
1. Member of *Bullied into Bad Science* campaign
2. Affiliate editor of *bioRxiv*
3. Senior editor of *Scientific Data*

Acknowledgements

Daniel Nust and the Mozilla foundation for https://codecheck.org.uk

Problems

Science as inverse problem



How do journals host code?

Eiko Fried
@EikoFried

Journal didn't know what to do with our #rstats supplementary .. so they took screenshots and pasted into word o.O

link.springer.com/article/10.100...

Electronic Supplementary Material

```
##########################################################
#                                                        #
#                 Supplementary Materials                #
#                                                        #
#          Mental disorders as networks of problems:     #
#                 a review of recent insights            #
#                                                        #
#                Fried, van Borkulo, et al., 2016        #
#                                                        #
#        Social Psychiatry and Psychiatric Epidemiology  #
#                                                        #
##########################################################

library(qgraph)

##### Figure 1

### Figure 1A
AdjacencyMatrix.1A <- matrix(0,8,8)
AdjacencyMatrix.1A[1,2] <- AdjacencyMatrix.1A[2,1] <-
AdjacencyMatrix.1A[2,3] <- AdjacencyMatrix.1A[3,2] <-
AdjacencyMatrix.1A[2,4] <- AdjacencyMatrix.1A[4,2] <-
AdjacencyMatrix.1A[2,5] <- AdjacencyMatrix.1A[5,2] <-
AdjacencyMatrix.1A[3,4] <- AdjacencyMatrix.1A[4,3] <-
AdjacencyMatrix.1A[3,5] <- AdjacencyMatrix.1A[5,3] <-
AdjacencyMatrix.1A[3,6] <- AdjacencyMatrix.1A[6,3] <-
AdjacencyMatrix.1A[3,8] <- AdjacencyMatrix.1A[8,3] <-
AdjacencyMatrix.1A[4,7] <- AdjacencyMatrix.1A[7,4] <-
AdjacencyMatrix.1A[5,7] <- AdjacencyMatrix.1A[7,5] <- 1
N.1A <- qgraph(AdjacencyMatrix.1A, border.width=2,edge.width=1.2,
vsize=10,
             border.color='#555555', edge.color="#555555",
label.color="#555555", color="#EEEEEE")

pdf("Figure1A.pdf")
qgraph(N.1A)
dev.off()

### Figure 1B
AdjacencyMatrix.1B <- matrix(0,8,8)
AdjacencyMatrix.1B[1,3] <- AdjacencyMatrix.1B[2,1] <-
AdjacencyMatrix.1B[2,5] <- AdjacencyMatrix.1B[2,3] <-
AdjacencyMatrix.1B[2,6] <- AdjacencyMatrix.1B[2,4] <-
AdjacencyMatrix.1B[2,8] <- AdjacencyMatrix.1B[8,2] <-
AdjacencyMatrix.1B[3,2] <- AdjacencyMatrix.1B[3,6] <-
AdjacencyMatrix.1B[4,8] <- AdjacencyMatrix.1B[4,7] <-
AdjacencyMatrix.1B[5,1] <- AdjacencyMatrix.1B[5,5] <-
AdjacencyMatrix.1B[8,8] <- AdjacencyMatrix.1B[7,7] <- 1
N.1B <- qgraph(AdjacencyMatrix.1B, border.width=2, edge.width=2,
```

11:52 PM · Dec 7, 2016 · Twitter Web Client

eLife: clone github repos (good), but also 'curate' to archive, changing filenames (bad).

https://twitter.com/eikofried/status/806647708853346306

Data accessibility statements

Many journals still allow the famous sentence:

> Code/data available upon REASONABLE request

Journals need to tighten up on this.

Peer review

Peer reviewers don't have time to look at code. (Or at least I don't).

I only ever received one review to date where a referee checked (and edited) my code.

We have "data papers"; what about "code papers"?

Few journals that want to take your purely code papers. Not seen as research.

Good neuro data repos still in infancy (perhaps with exception of neuroimaging) to support journal policy

Some journals sharing "workflows" (Gigascience, Scientific Data).

Opportunities

Sharing code now seen as important

- Saga of Imperial College (UK) COVID model has shown that finally code seen as important. People want to see the code of the model. (That has been used since early 2000s).

https://www.nature.com/articles/d41586-020-01003-6

Journal initiatives

- Growing appetite among journals with checks of code.
- PLOS Comp Biol have launched a code review May 2020.
- Spectrum of reproducibility: "reproducible forever, by everyone" (CodeOcean), to "it worked once for someone else outside the group" (CODECHECK).
- example certificate: https://zenodo.org/record/3674056

Other issues that impact on journals.

- Funders are asking for other outputs, beyond traditional paper publication.
- Likewise (in UK) research outputs can include research artifacts. And what the funders ask for, journals I hope will help provide.
- Plan S: will helping curate your code be something that journals might use to justify APC? (Just like with a data paper, like at Scientific Data [disclaimer: am editor there]
- Rise in the "research software engineers" that are helping scientists write better code that we are happier to share.
- small research pots available for open science, hopefully they will grow.